Tanishq Rajiv Bhujbale

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.12
Sant Tukaram National Model School Latur	Class XII (CBSE)	2024	90.80%
Sharada International School Latur	Class X (CBSE)	2022	92.20%

SKILLS

Programming Languages : C, C++, Python

Frameworks & Libraries : NumPy, Pandas, Matplotlib, Pytorch

Databases : MySQL, PostgreSQL

Developer Tools : Git, GitHub, Linux, Latex,

Conecpts : Data Structures & Algorithms, Machine Learning, Deep Learning

RELEVANT COURSES

Mathematics : Discrete Maths, Matrix Theory, Probability & Random Variables, Convex Optimization, Linear System and Signal Processing, Control System

Programming : Introduction to Programming, Foundations of Machine Learning, Data Structures & Application, Algorithms, Deep Learning

PROJECTS

Bayesian & Linear Regression Framework

- Built Polynomial Regression (Gradient Descent & Normal Equation) and Bayesian Linear Regression with Gaussian basis functions from scratch using only NumPy, without machine learning libraries.

- Implemented sequential Bayesian posterior updates, MAP estimation, posterior predictive inference, and regression curve sampling while thoroughly analyzing convergence behavior, structural uncertainty, L2 regularization, and the fundamental bias-variance trade-off through extensive empirical experimentation and comprehensive data visualization.

Deep Sequence Modeling for Sentiment Classification

- Implemented Vanilla RNN, LSTM, Transformer Encoder, and Selective State Space Models (SSMs) entirely from scratch using NumPy to perform highly accurate binary sentiment classification on the large-scale IMDb movie review dataset.

- Successfully developed customized word embeddings, Backpropagation Through Time (BPTT), gradient clipping thresholds, multi-head self-attention mechanisms, sinusoidal positional encoding vectors, residual connections, and layer norm layers.

- Compared recurrent, attention-based, and state-space architectures by conducting rigorous empirical ablation studies across variable sequence lengths scaling up to 1024 distinct tokens.

CNN-Based Image Classification

- Built a scratch NumPy CNN (conv, pooling, activation, FC layers); trained on CIFAR-10 using PyTorch.

- Evaluated optimization methods (SGD, Momentum, RMSProp) and regularization techniques (Dropout, Batch Normalization) through detailed analysis of accuracy, loss curves, activation maps, and high-dimensional t-SNE embeddings.

Neural Network Framework from Scratch

- Built a modular deep learning framework in NumPy featuring ADALINE and configurable MLPs with custom activation functions, weight initialization methods, and optimizers (SGD, Momentum, RMSProp, Adam).

- Developed a synthetic 5,000-sample IIT-H Campus Life predictive dataset and successfully integrated L1/L2 weight regularization penalties, early stopping criteria, and a highly modular, flexible experimentation pipeline.

- Evaluated structural architectural variations and optimization design choices through comprehensive hyperparameter studies focusing on network depth, layer width, activation functions, regularization strengths, and optimizer performance, leveraging training convergence metrics, validation performance indicators, and learned feature visualizations.

Unsupervised Learning and Dimensionality Reduction

- Implemented Principal Component Analysis (PCA) completely from scratch for facial recognition pipelines, including detailed components for structural dimensionality reduction, eigenvalues calculation, image reconstruction, and comprehensive reconstruction error analysis.

- Developed K-Means and Gaussian Mixture Models (GMMs) using the Expectation-Maximization (EM) algorithm, comparing clustering performance, convergence behavior, and visualization across multiple synthetic datasets.

EXTRACURRICULAR

BGMI : Secured **1st Place** in Inter Hostel Competetion at IIT Hyderabad

Geoguessr : Ranked top **50** in India as of June 2026